Equations_Snippets — Node Documentation for SDF-Lattice Core Nodes (Permanent Record)

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1 Purpose and Persistence Mandate

This document provides the permanent, node-level documentation for the SDF-Lattice core node chain, written to be archived inside the `Equations_Snippets` folder of the `Paper-SDF-NGC6503` repository. Its purpose is to prevent "evaporation" of the derivations: every equation, constant, validation threshold, and provenance anchor recorded here is taken verbatim from the canonical codebase (`sdf_lattice/nodes/` in `SDF_Lattice_Master_v0.3.3`, node modules `strain.py`, `permittivity.py`, `void_coupling.py`, `closure_constraints.py`) and from snippets 01–04 (all last modified 2026-11-01). The snippets themselves are treated strictly as phenomenological effective Ansätze; their justification flows upstream from the axiomatic layer in `SDF-VLT-Gravity-Dynamics`. Any future edit to a node must be accompanied by a corresponding edit here, so that the recorded mathematics never dangles without a physical source of truth.

2 Canonical Node Chain and Dataflow

The canonical registration order, read directly from the package `__init__.py`, is:

```
Node → PassthroughNode → StrainTensorNode → PermittivityEnvelopeNode
   → VoidCouplingNode → ClosureConstraintsNode → SupercellCouplingNode
   → RetardedCausalityNode → BerryDynamicsNode → BerryTorqueNode
   → PeristalticPumpNode → CavityResilienceNode
```

Dataflow is enforced by each node's `required_inputs`. The validated physics pipeline is:

StrainTensorNode $\rightarrow$ PermittivityEnvelopeNode $\rightarrow$ VoidCouplingNode $\rightarrow$ ClosureConstraintsNode $\rightarrow$ SupercellCouplingNod

In words: the strain tensor feeds the permittivity envelope; the envelope's `radiative_flux` feeds void coupling; void coupling's `phase_coherence` feeds the closure constraints, which decide stability before any supercell aggregation. Every quantity documented below is produced at one link of this chain and consumed at the next.

3 StrainTensorNode and PermittivityEnvelopeNode

StrainTensorNode (`strain.py`, 101 lines). Inputs: `tensor` or `displacement_gradient` — a symmetric, finite 2×2 or 3×3 matrix; symmetry violations raise `"Strain tensor must be symmetric"`. Equations exactly as implemented:

$$\varepsilon = \tfrac{1}{2}\left(\text{data} + \text{data}^{\mathsf{T}}\right), \qquad \text{trace} = \operatorname{tr}\varepsilon$$

$$\varepsilon_{\text{hydro}} = \frac{\operatorname{tr}\varepsilon}{d}, \qquad \varepsilon_{\text{dev}} = \varepsilon - \varepsilon_{\text{hydro}}\, I, \qquad \|\varepsilon_{\text{dev}}\|_F = \|\varepsilon_{\text{dev}}\|_{\text{Frobenius}}$$

Criticality flag: `is_critical` is set when $\|\varepsilon_{\text{dev}}\|_F \geq$ `critical_threshold` (class default **1.0**). A guard comment at L76–77 is normative: "Strain remains dimensionless and is never reinterpreted as stress." Stress is emitted only when an explicit Berry gap stress is supplied (`berry_gap_stress` or `berry_torque_result.gap_stress`), giving `effective_gap_stress = base_gap_stress + berry_stress`.

PermittivityEnvelopeNode (`permittivity.py`, 87 lines). Fixed parameters: `epsilon_0 = 1.0`, `photoelastic_coupling = 0.15`, `radiative_leak_rate = 0.85` (must lie in [0, 1]). Required inputs are exactly the outputs of the strain node (`trace`, `frobenius_dev`, `is_critical`). Equations exactly as implemented (L76–78):

$$\varepsilon_{\text{eff}} = \varepsilon_0 \left(1 + 0.15 \cdot \operatorname{tr}\varepsilon\right)$$

$$\rho_{\text{reactive}} = \tfrac{1}{2}\, \varepsilon_{\text{eff}} \left((\operatorname{tr}\varepsilon)^2 + \|\varepsilon_{\text{dev}}\|_F^2 \right)$$

$$\Phi_{\text{rad}} = \begin{cases} 0.85 \cdot \rho_{\text{reactive}} & \text{if critical} \\ 0 & \text{otherwise} \end{cases}$$

Output `confinement_status` ∈ `{"radiative", "confined"}` mirrors `is_critical`. The photoelastic constant 0.15 is the code-level strain-to-permittivity bridge; it is the microscopic counterpart of the macroscopic foam coupling α_{foam} in snippet 02.

4 VoidCouplingNode: Five-Edge Phaseors and Phase Locking

VoidCouplingNode (`void_coupling.py`, 109 lines). Parameters: `coupling_constant = 1.0` (> 0), `critical_coherence_threshold = 0.7` (must lie in [0, 1]), `bridge_phase_offset = 0.0`. Input `edge_phases` must contain exactly 5 phases — the five-edge arrangement of the pentatetrahedral cavity — plus optional `radiative_flux` (non-negative) from the permittivity node and `vector_potential_samples` (A_i, dl_i). Equations exactly as implemented (L88–100):

$$\text{re} = \frac{1}{5}\sum_{i=1}^{5} \cos\theta_i, \qquad \text{im} = \frac{1}{5}\sum_{i=1}^{5} \sin\theta_i$$

$$\text{phase_coherence} = \text{clamp}\left(\sqrt{\text{re}^2 + \text{im}^2},\ 0,\ 1\right), \qquad \text{mean_phase} = \text{atan2}(\text{im}, \text{re}) \ \ (\text{zero if coherence} \leq 10^{-15})$$

$$\text{bridge_phase} = (\text{mean_phase} + \text{offset}) \bmod 2\pi$$

$$E_{\text{coupling}} = \tfrac{1}{2}\, g \cdot \text{coherence}^2 \cdot (1 + \Phi_{\text{rad}}), \qquad \Phi_{\text{cavity}} = \textstyle\sum_i A_i\, dl_i \quad (\text{via } \texttt{math.fsum})$$

$$C_{jk} = \cos(\theta_j - \theta_k) \quad (5\times5 \text{ coupling matrix})$$

Phase lock: `is_phase_locked` holds whenever `coherence` ≥ `0.7`. Geometric provenance: the five-edge count comes from the pentatetrahedral cavity arrangement, whose Euclidean non-closure is set by the tetrahedral angle deficit documented in `docs/5_tetrahedron_cavity_whitepaper.en.md`:

$$\Delta\theta = 2\pi - 5\arccos\left(\tfrac{1}{3}\right) \approx 7.36^\circ$$

This deficit is the structural origin of the cavity's residual curvature and of the phase mismatch the node measures. Loss of coherence corresponds physically to the breaking of void chains, after which gravity emerges residually as $\Phi_{\text{grav}} = -G_{\text{eff}} M_{\text{residual}}/r$ (whitepaper L143–145).

5 ClosureConstraintsNode and Stability State Machine

ClosureConstraintsNode (`closure_constraints.py`, 93 lines). Parameters: `causality_threshold = 1.50`, `energy_threshold = 0.50`, `min_coherence_threshold = 0.20`. Inputs: `strain_tensor`, `permittivity_envelope`, `coupling_strength` (with alternates `strain/strain_matrix`, `permittivity/envelope`, `coupling/g_coupling`), and optional `phase_coherence` imported from VoidCouplingNode (explicit at L49). Equations exactly as implemented (L59–65):

$$\langle \varepsilon^2 \rangle = \text{mean of all squared strain entries}, \qquad \text{Var}(\varepsilon) = \text{envelope dispersion}, \qquad \bar{\varepsilon} = \text{mean envelope value}$$

$$\text{causality_residual} = \left| \langle \varepsilon^2 \rangle - g\bar{\varepsilon} - \text{Var}(\varepsilon) \right|$$

$$\text{energy_deficit} = \left| 1 - \bar{\varepsilon}\left(1 + 0.1\,\langle \varepsilon^2 \rangle\right) \right|$$

Stability rule (L68–86):

$$\text{is_stable} = (\text{causality_residual} < 1.50) \wedge (\text{energy_deficit} < 0.50) \wedge (\text{coherence} \geq 0.20)$$

The closure state machine emits exactly four states: `BOUND_STABLE` (all three conditions met), `BIFURCATION_COLLAPSE`, `PHASE_LEAKAGE`, and `ENERGY_DIVERGENCE` (each corresponding to which constraint is violated). These residuals are the code-level expression of the pre-Friedmann closure constraints stated in the tetrahedral cavity whitepaper (L1). Outputs: `causality_residual`, `energy_deficit`, `closure_status`, `is_stable`.

6 Semantic Mapping: Code Quantities to Equations_Snippets 01–04

The four snippets are phenomenological Ansätze; each macroscopic quantity has a microscopic counterpart in the node chain. The mapping below must accompany any snippet edit.

Code quantity (node)	Physical counterpart in snippets
`phase_coherence` (VoidCoupling)	Foam order parameter coherence; its loss = broken void chains → emergent $\Phi_{\text{grav}} = -G_{\text{eff}} M_{\text{residual}}/r$
`frobenius_dev` + `critical_threshold`	Activation of shear modes → source of anisotropic $\Pi_{\mu\nu}$ (snippet 01)
`photoelastic_coupling = 0.15`	Microscopic strain → permittivity bridge → macroscopic coupling α_{foam} (snippet 02)
`radiative_flux` (PermittivityEnvelope)	Enters E_{coupling} as $(1 + \Phi_{\text{rad}})$ — radiative leakage of confined vacuum modes
`causality_residual`, `energy_deficit` (ClosureConstraints)	Pre-Friedmann closure constraints of the cavity whitepaper
$a_0 \approx 1.2 \times 10^{-10}\,\text{m/s}^2$ (snippet 02)	Anchored to the VLT root relation $a_0 \equiv 2\pi G\,\sigma_{\text{wall}} = cH_0/2\pi$
$v_\infty \approx 116.2\,\text{km/s}$ (snippet 03)	Empirical asymptotic saturation for NGC 6503, matched across archives
$\zeta(r)$ unfreezing profile (snippet 03)	To be superseded by the formal profile from `SDF-VLT/02_Boundary_Pressure_Gravity/04_Cavity_Boundary_Intera` the ad-hoc `tanh` in `radial_unfreezing_verification.py` is deprecated

Snippet-to-axiom anchoring plan (repair of the README dead links): snippet 01 imports `SDF-VLT-Gravity-Dynamics/02_Bounda` snippet 02 anchors a_0 to `04_Observational_Validation/01_Galactic_Rotation_Curves.md` (equation $a_0 \equiv 2\pi G\sigma_{\text{wall}}$); snippet 04 anchors S_E to `00_Root_Governance/NT_ECS_Noether_Conservation.md`. Two README references remain unresolved and are hereby registered as pending: snippet 05 (`05_ngc6503_metric_perturbations.md`, formula $\eta(r) = \Psi/\Phi = 1 + \Delta_{\text{defect}}$, listed but the file is absent from these archives) and the script `nodes/active_tensor_foam.py` (conceptual equivalents exist in the VLT geometric-emergence nodes; a dedicated mapping file must be authored to close these links). Recording this mapping inside `Equations_Snippets` guarantees the provenance chain survives any future archive migration.